



Module 1: Cloud Concepts Overview

AWS Academy Cloud Foundations

© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Welcome to Module 1: Cloud Concepts Overview.

Section 1: Introduction to cloud computing

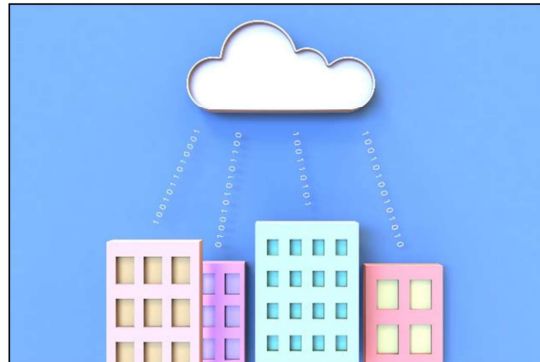
Module 1: Cloud Concepts Overview



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Section 1: Introduction to cloud computing

What is cloud computing?

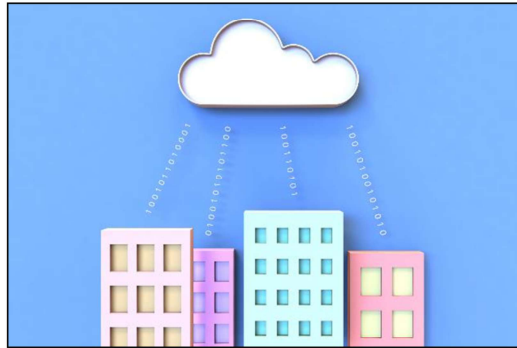


What does cloud computing mean to you?

Take a moment to think of what cloud computing means to you and write a short sentence.

Cloud computing defined

Cloud computing is the **on-demand** delivery of compute power, database, storage, applications, and other IT resources **via the internet** with **pay-as-you-go** pricing.



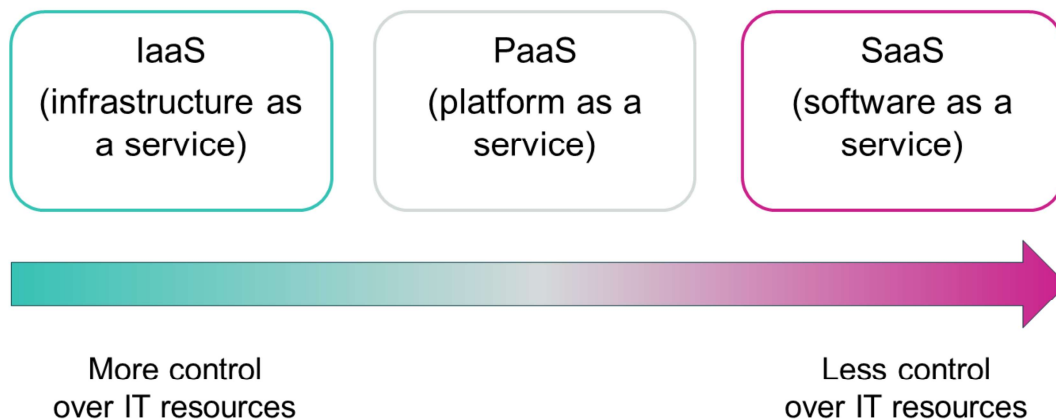
© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

4

Cloud computing is the on-demand delivery of compute power, database, storage, applications, and other IT resources via the internet with pay-as-you-go pricing. These resources run on server computers that are located in large data centers in different locations around the world. When you use a cloud service provider like AWS, that service provider owns the computers that you are using. These resources can be used together like building blocks to build solutions that help meet business goals and satisfy technology requirements.

To learn more about cloud computing and how it works, see <https://aws.amazon.com/what-is-cloud-computing/>.

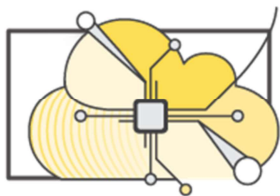
Cloud service models



There are three main cloud service models. Each model represents a different part of the cloud computing stack and gives you a different level of control over your IT resources:

- **Infrastructure as a service (IaaS):** Services in this category are the basic building blocks for cloud IT and typically provide you with access to networking features, computers (virtual or on dedicated hardware), and data storage space. IaaS provides you with the highest level of flexibility and management control over your IT resources. It is the most similar to existing IT resources that many IT departments and developers are familiar with today.
- **Platform as a service (PaaS):** Services in this category reduce the need for you to manage the underlying infrastructure (usually hardware and operating systems) and enable you to focus on the deployment and management of your applications.
- **Software as a service (SaaS):** Services in this category provide you with a completed product that the service provider runs and manages. In most cases, software as a service refers to end-user applications. With a SaaS offering, you do not have to think about how the service is maintained or how the underlying infrastructure is managed. You need to think only about how you plan to use that particular piece of software. A common example of a SaaS application is web-based email, where you can send and receive email without managing feature additions to the email product or maintaining the servers and operating systems that the email program runs on.

Cloud computing deployment models



Cloud



Hybrid



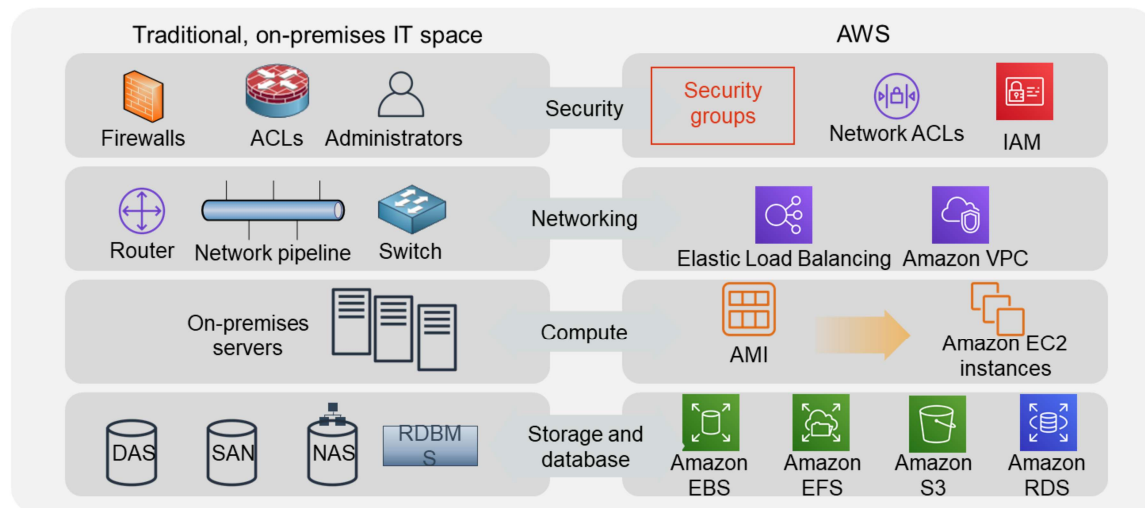
On-premises
(private cloud)



There are three main cloud computing deployment models, which represent the cloud environments that your applications can be deployed in:

- **Cloud:** A cloud-based application is fully deployed in the cloud, and all parts of the application run in the cloud. Applications in the cloud have either been created in the cloud or have been migrated from an existing infrastructure to take advantage of the benefits of cloud computing (see <https://aws.amazon.com/what-is-cloud-computing/>). Cloud-based applications can be built on low-level infrastructure pieces or they can use higher-level services that provide abstraction from the management, architecting, and scaling requirements of core infrastructure.
- **Hybrid:** A hybrid deployment is a way to connect infrastructure and applications between cloud-based resources and existing resources that are not located in the cloud. The most common method of hybrid deployment is between the cloud and existing on-premises infrastructure. This model enables an organization to extend and grow their infrastructure into the cloud while connecting cloud resources to internal systems.
- **On-premises:** Deploying resources on-premises, using virtualization and resource management tools, is sometimes called private cloud. While on-premises deployment does not provide many of the benefits of cloud computing, it is sometimes sought for its ability to provide dedicated resources. In most cases, this deployment model is the same as legacy IT infrastructure, but it might also use application management and virtualization technologies to increase resource utilization.

Similarities between AWS and traditional IT



There are many similarities between AWS and the traditional, on-premises IT space:

- AWS security groups, network access control lists (network ACLs), and AWS Identity and Access Management (IAM) are similar to firewalls, access control lists (ACLs), and administrators.
- Elastic Load Balancing and Amazon Virtual Private Cloud (Amazon VPC) are similar to routers, network pipelines, and switches.
- Amazon Machine Images (AMIs) and Amazon Elastic Compute Cloud (Amazon EC2) instances are similar to on-premises servers.
- Amazon Elastic Block Store (Amazon EBS), Amazon Elastic File System (Amazon EFS), Amazon Simple Storage Service (Amazon S3), and Amazon Relational Database Service (Amazon RDS) are similar to direct attached storage (DAS), storage area networks (SAN), network attached storage (NAS), and a relational database management service (RDBMS).

With AWS services and features, you can do almost everything that you would want to do with a traditional data center.

Section 2: Advantages of cloud computing

Module 1: Cloud Concepts Overview



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Section 2: Advantages of cloud computing

Why are so many companies interested in moving to the cloud? This section presents six advantages of cloud computing.

Trade capital expense for variable expense



Data center investment
based on forecast



Pay only for the amount
you consume



Advantage #1—Trade capital expense for variable expense: Capital expenses (capex) are funds that a company uses to acquire, upgrade, and maintain physical assets such as property, industrial buildings, or equipment. Do you remember the data center example in the traditional computing model where you needed to rack and stack the hardware, and then manage it all? You must pay for everything in the data center whether you use it or not.

By contrast, a *variable expense* is an expense that the person who bears the cost can easily alter or avoid. Instead of investing heavily in data centers and servers before you know how you will use them, you can pay only when you consume resources and pay only for the amount you consume. Thus, you save money on technology. It also enables you to adapt to new applications with as much space as you need in minutes, instead of weeks or days. Maintenance is reduced, so you can spend focus more on the core goals of your business.

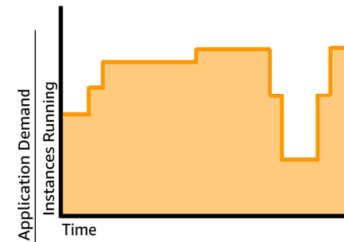
Stop guessing capacity



Overestimated
server capacity



Underestimated
server capacity



Scaling on
demand

Advantage #3—Stop guessing capacity: Eliminate guessing about your infrastructure capacity needs. When you make a capacity decision before you deploy an application, you often either have expensive idle resources or deal with limited capacity. With cloud computing, these problems go away. You can access as much or as little as you need, and scale up and down as required with only a few minutes' notice.

Increase speed and agility



Weeks between wanting
resources and having resources

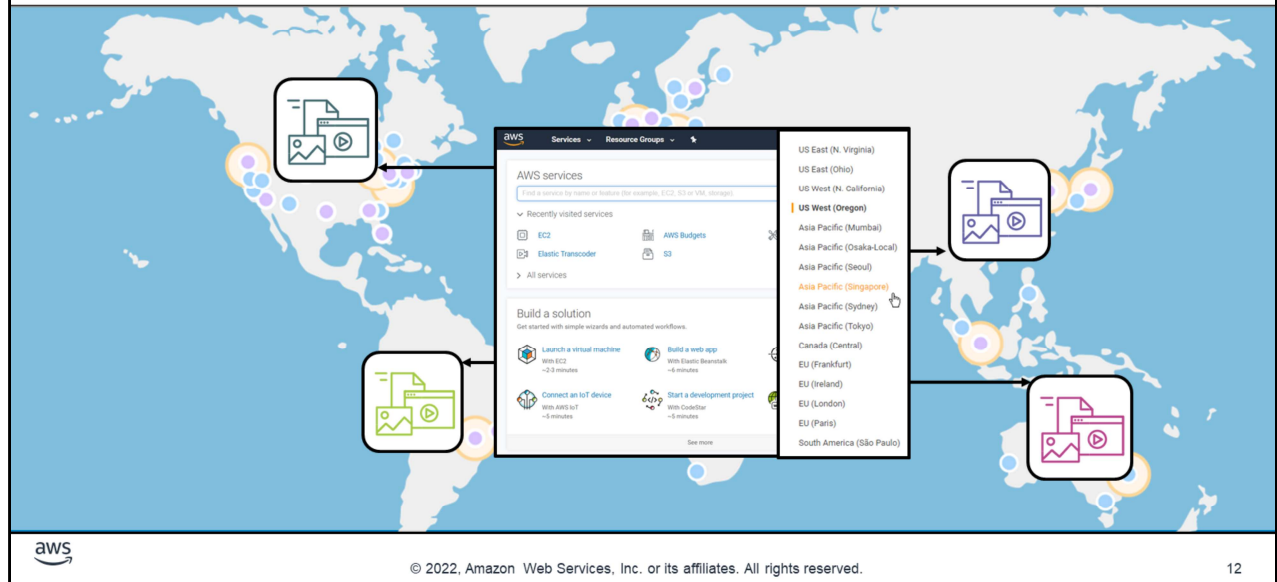


Minutes between wanting
resources and having
resources



Advantage #4—Increase speed and agility: In a cloud computing environment, new IT resources are only a click away, which means that you reduce the time it takes to make those resources available to your developers from weeks to just minutes. The result is a dramatic increase in agility for the organization because the cost and time that it takes to experiment and develop are significantly lower.

Go global in minutes



Advantage #6—Go global in minutes: You can deploy your application in multiple AWS Regions around the world with just a few clicks. As a result, you can provide a lower latency and better experience for your customers simply and at minimal cost.

Section 3: Introduction to Amazon Web Services (AWS)

Module 1: Cloud Concepts Overview

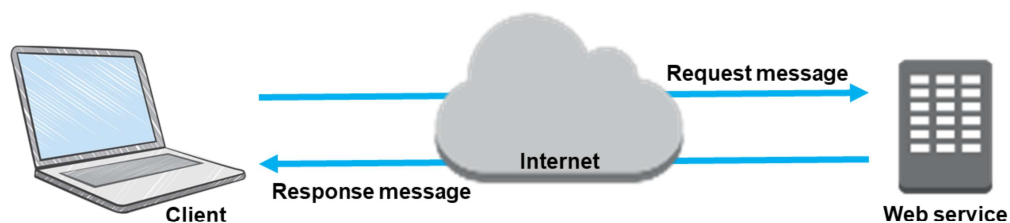


© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Section 3: Introduction to Amazon Web Services (AWS)

What are web services?

A **web service** is any piece of software that makes itself available over the internet and uses a **standardized format**—such as Extensible Markup Language (XML) or JavaScript Object Notation (JSON)—for the request and the response of an **application programming interface (API) interaction**.



In general, a web service is any piece of software that makes itself available over the internet or on private (intranet) networks. A web service uses a standardized format—such as Extensible Markup Language (XML) or JavaScript Object Notation (JSON)—for the request and the response of an application programming interface (API) interaction. It is not tied to any one operating system or programming language. It's self-describing via an interface definition file and it is discoverable.

What is AWS?

- AWS is a **secure cloud platform** that offers a **broad set of global cloud-based products**.
- AWS provides you with **on-demand access** to compute, storage, network, database, and other IT resources and management tools.
- AWS offers **flexibility**.
- You **pay only for the individual services you need, for as long as you use them**.
- AWS services **work together** like building blocks.



Amazon Web Services (AWS) is a secure cloud platform that offers a broad set of global cloud-based products. Because these products are delivered over the internet, you have on-demand access to the compute, storage, network, database, and other IT resources that you might need for your projects—and the tools to manage them. You can immediately provision and launch AWS resources. The resources are ready for you to use in minutes.

AWS offers flexibility. Your AWS environment can be reconfigured and updated on demand, scaled up or down automatically to meet usage patterns and optimize spending, or shut down temporarily or permanently. The billing for AWS services becomes an operational expense instead of a capital expense.

AWS services are designed to work together to support virtually any type of application or workload. Think of these services like building blocks, which you can assemble quickly to build sophisticated, scalable solutions, and then adjust them as your needs change.

Categories of AWS services

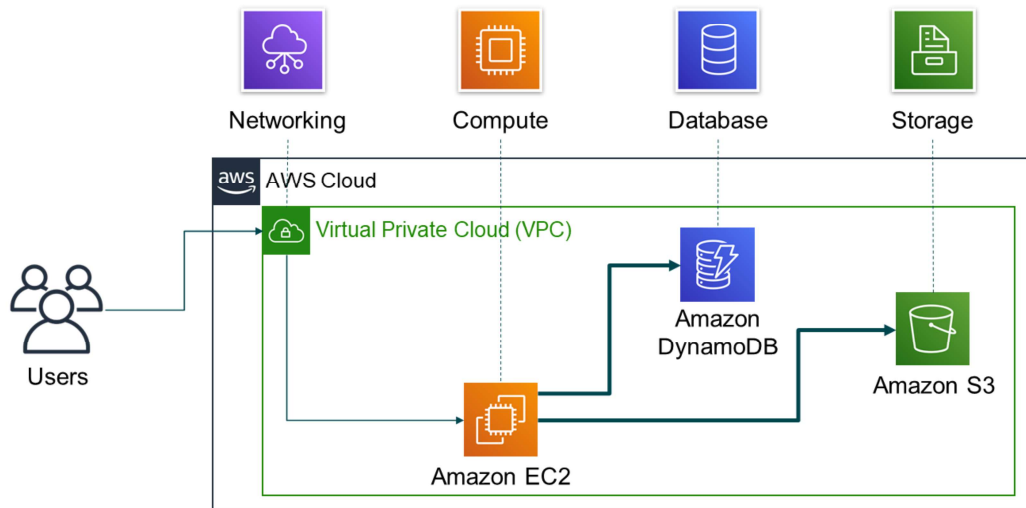


© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

16

AWS services fall under different categories, and each category contains one or more services. You can select the services that you want from these different categories to build your solutions.

Simple solution example



For example, say you're building a database application. Your customers might be sending data to your Amazon Elastic Compute Cloud (Amazon EC2) instances, which is a service in the compute category. These EC2 servers batch the data in one-minute increments and add an object per customer to Amazon Simple Storage Service (Amazon S3), the AWS storage service you've chosen to use. You can then use a nonrelational database like Amazon DynamoDB to power your application, for example, to build an index so that you can find all the objects for a given customer that were collected over a certain period. You might decide to run these services inside an Amazon Virtual Private Cloud (Amazon VPC), which is a service in the networking category.

The purpose of this simple example is to illustrate that you can select web services from different categories and use them together to build a solution (in this case, a database application). Of course, the solutions you build can be quite complex.

Three ways to interact with AWS



AWS Management Console

Easy to use graphical interface



Command Line Interface (AWS CLI)

Access to services by discrete commands or scripts



Software Development Kits (SDKs)

Access services directly from your code (such as Java, Python, and others)



You might wonder how to access the broad array of services that are offered by AWS. There are three ways to create and manage resources on the AWS Cloud:

- **AWS Management Console:** The console provides a rich graphical interface to a majority of the features offered by AWS. (Note: From time to time, new features might not have all of their capabilities included in the console when the feature initially launches.)
- **AWS Command Line Interface (AWS CLI):** The AWS CLI provides a suite of utilities that can be launched from a command script in Linux, macOS, or Microsoft Windows.
- **Software development kits (SDKs):** AWS provides packages that enable accessing AWS in a variety of popular programming languages. This makes it easy to use AWS in your existing applications and it also enables you to create applications that deploy and monitor complex systems entirely through code.

All three options are built on a common REST-like API that serves as the foundation of AWS.

To learn more about tools you can use to develop and manage applications on AWS, see <https://aws.amazon.com/tools/>.